Name of the Student	N.AKSHAYA
Reg. No	192425129
GitHub Link	https://github.com/akshaya24032007/MLA0405-DeepLearning/tree/main/Assessment/CO5

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: **Deep Learning** **COURSE CODE:** **MLA0405**
SLOT : D

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35% **Duration: 90 minutes**
Assessment Tool Weightage: 20% **Mark : 25**

Code of Conduct:
I **N.AKSHAYA(192425129)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.AKSHAYA **Name of the student:** N.AKSHAYA

Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking **(2)**	Communication Skills and Clarity of Explanation **(1.5)**	Confidence, Presentation and Professional Behaviour ***(1)***	**Total (10)**
Weightage	***30 %***	***25%***	***20%***	***15 %***	***10 %***	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						

Q9						
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Q19						
Q20						
Q21						
Q22						
Q23						
Q24						
Q25						

Signature of the Faculty: **Total Marks:**

QUESTIONS: 25 **Total Marks:25**

1.What is an Autoencoder in Deep Learning?
Answer:
A neural network that learns to reconstruct input data.
2.What is the main purpose of an Autoencoder?
Answer:
Data reconstruction and feature learning.
3.What are the major components of an Autoencoder?
Answer:
Components: Encoder, bottleneck, and decoder.
4.What is the role of the Encoder in an Autoencoder?
Answer:
Converts input into latent representation.
5.What is the role of the Decoder in an Autoencoder?
Answer:
Reconstructs the original input.
6.What is meant by the Bottleneck layer in an Autoencoder?
Answer:
Compressed latent representation.
7.What is meant by Reconstruction in an Autoencoder?
Answer:
Reproducing the original input.
8.How does an Autoencoder learn to reconstruct its input?
Answer:
By minimizing reconstruction loss.
9.What is the difference between an Autoencoder and a traditional Neural Network?
Answer:

Autoencoder is mainly unsupervised.
10.What type of loss function is commonly used to train an Autoencoder?
Answer:
Mean Squared Error (MSE).
11.Explain the basic architecture of an Autoencoder with a suitable example.
Answer:
Input → Encoder → Bottleneck → Decoder → Output.
12.Why is the dimensionality of the bottleneck layer usually smaller than the input layer?
Answer:
To achieve compression.
13.What happens if the bottleneck layer has too many neurons?
Answer:
Poor compression and overfitting.
14.What happens if the bottleneck layer has very few neurons?
Answer:
Information loss.
15.What is a Denoising Autoencoder?
Answer:
Removes noise from corrupted inputs.
16.How can an Autoencoder be used for image denoising?
Answer:
Learns to reconstruct clean images from noisy images.
17.How can Autoencoders be used for dimensionality reduction?
Answer:
Uses the bottleneck representation.
18.How can an Autoencoder be used for anomaly detection?
Answer:
Detects high reconstruction error.
19.What is the difference between a Denoising Autoencoder and a basic Autoencoder?
Answer:
Denoising Autoencoder is trained with noisy inputs.
20.Give some real-world applications of Autoencoders.
Answer:
Denoising, anomaly detection, and compression.
21.What is a Deep Generative Model?
Answer:
Generates new data similar to training data.
22.What is the difference between a generative model and a discriminative model?
Answer:
Generative models create data; discriminative models classify data.
23.What is a Variational Autoencoder (VAE)?
Answer:
A probabilistic generative autoencoder.
24.What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?
Answer:
VAE learns a probability distribution; traditional AE learns representations.
25.What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?
Answer:
Uses a Generator to create data and a Discriminator to distinguish real from fake.

PHOTO:

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation and Professional Behaviour	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis